Engineering Analysis Report

Furnace Oxide Thickness: Statistical Process Control and Process Capability

Exploratory analysis of the NIST/SEMATECH furnace thickness dataset | Portfolio study

1. Objective and scope

Evaluate measured silicon oxide film thickness against a target of 560 Å and specification limits of 460–660 Å. Examine differences among furnace runs, furnace zones and paired wafer measurements, and identify priorities for process investigation. This is an observational study; operating conditions and wafer lot information were not provided.

2. Data and methodology

The dataset contains 168 thickness measurements: 21 furnace runs × 4 zones × 2 sampled wafers per run–zone subgroup. The analysis includes descriptive statistics, run/zone visualisations, provisional Shewhart X-bar and R charts (subgroup size 2), exploratory two-factor ANOVA with interaction, and capability/performance indices.

Metric	Observed value
Target / specifications	560 Å / 460–660 Å
Sample size	168 observations
Observed mean / standard deviation	563.04 Å / 25.38 Å
Observed minimum / maximum	487 Å / 634 Å

3. Statistical process control

The X-bar charts show multiple points beyond provisional control limits in all four zones. R charts show fewer such signals; Zone 1, Run 20 has a wafer-pair range of 40 Å, exceeding its upper control limit (approximately 37.96 Å). The notebook records a suspected additional Zone 2 R-chart signal, but exact counts for Zones 2–4 should be verified numerically before final publication. Limits were estimated using the same 21 runs under study (Phase I); subgroup suitability and sampling assumptions remain unverified.

Mean wafer-pair ranges by zone were approximately 11.62, 12.14, 14.86 and 8.33 Å (Zones 1–4). Relatively close measurements within a pair can coexist with large shifts in subgroup averages between runs.

4. Exploratory ANOVA

A fixed-effects model with RUN, ZONE and RUN × ZONE interaction yielded statistically significant RUN and interaction terms (p approximately 1.89 × 10^-27 and 2.95 × 10^-11 respectively). The overall ZONE term had p approximately 0.0637, above the conventional 0.05 threshold. With an interaction present, the main ZONE effect should not be interpreted in isolation. Independence and error-model assumptions require validation; statistical significance does not identify a physical cause.

5. Capability and performance

Index	Estimate	Basis
Cp	3.203	Within-pair variation
Cpk	3.106	Within-pair variation and centering
Pp	1.313	Overall standard deviation
Ppk	1.273	Overall standard deviation and centering

The gap between Cp/Cpk and Pp/Ppk indicates substantially more dispersion across the full dataset than within paired wafers from the same run and zone. All observed measurements met specifications, but out-of-control chart signals prevent a defensible claim of demonstrated stable process capability or future yield.

6. Engineering conclusions

Principal finding: The sample meets the stated thickness specifications, yet statistical evidence points to changes in the process distribution across runs and changing zone patterns. Variation beyond the two-wafer subgroup warrants investigation. Temperature, oxidation time, gas environment, pressure, wafer material/lot effects and measurement-system factors are possible hypotheses, not established root causes.

7. Recommendations and limitations

For a production investigation: (1) validate the measurement system and subgroup selection; (2) examine out-of-control runs and obtain furnace temperature, time, gas flow/composition, pressure, recipe, maintenance and wafer lot records; (3) assess run and zone variance components with a design-appropriate model; (4) establish prospective control limits and reassess capability only after process stability is demonstrated.

Limitations include unavailable operating-parameter data, unknown wafer sourcing, only two wafers per run-zone subgroup, untested model assumptions and Phase I control limits. Results are exploratory and are not a formal process qualification.

8. Source and reproducibility

Source: NIST/SEMATECH Engineering Statistics Handbook, Furnace Case Study: https://www.itl.nist.gov/div898/handbook/ppc/section5/ppc51.htm . Calculations, plots and implementation details are in furnace_spc_analysis_annotated.ipynb. Original source file FURNACE.DAT must be placed in data/FURNACE.DAT to rerun the notebook.